Algebraic structures and group preliminaries

Rings, fields, vector spaces, subgroups, and cyclic groups

Applied Algebra
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Where we are, and where we are going

Last lecture focused on groups:

$$(\mathbb{Z}, +), \qquad (\mathbb{Z}_n, +), \qquad G_n, \qquad S_n, \qquad A_n, \qquad D_n.$$

Today we do two things.

Block	Topic
1	Rings and fields: two operations, zero-divisors, $\mathbb{Z}_p$.
2	Vector spaces: fields acting on Abelian groups.
3	Group preliminaries: cancellation, powers, subgroups, cyclic groups.

Rings and fields

From groups to rings

A group has **one operation**. A ring has **two operations**, usually

$$+ \quad \text{and} \quad \cdot\,.$$

The motivating example is

$$(\mathbb{Z}, +, \cdot).$$

Addition behaves like an Abelian group:

$$0, \quad -a, \qquad a + b = b + a.$$

Multiplication is associative and distributes over addition:

$$a(b + c) = ab + ac, \qquad (a + b)c = ac + bc.$$

Definition of a ring

A **ring** is a set R with two operations $+$ and $\cdot$ such that:

1. $(R, +)$ is an Abelian group; in particular, it has an additive identity 0;
2. multiplication is closed and associative;
3. multiplication distributes over addition:

$$x(y+z) = xy + xz, \qquad (x+y)z = xz + yz.$$

Useful vocabulary:

- an **identity element for multiplication** is an element $1 \in R$ such that

$$1a = a1 = a \qquad \text{for all } a \in R;$$

  a ring may or may not have such an element;
- a ring is **commutative** if $ab = ba$ for all $a, b \in R$.

Example: $\mathbb{Z}$ and $2\mathbb{Z}$

Example 1.

$$\mathbb{Z} = \{\dots, -2, -1, 0, 1, 2, \dots\}$$

is a commutative ring with identity 1.

Example 2.

$$2\mathbb{Z} = \{2k : k \in \mathbb{Z}\}$$

is also a ring under the usual operations.

But $2\mathbb{Z}$ has **no multiplicative identity**.

So a ring need not have all the familiar properties of $\mathbb{Z}$.

Example: matrix rings

Let

$$\mathrm{M}_2(\mathbb{R}) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} : a, b, c, d \in \mathbb{R} \right\}.$$

Under matrix addition and multiplication, $\mathrm{M}_2(\mathbb{R})$ is a ring with identity

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

But it is not commutative in general, and it has zero-divisors:

$$\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}.$$

$\mathbb{Z}_n$ as a ring

For $n \geq 2$, the congruence classes modulo n form a ring:

$$\mathbb{Z}_n = \{[0]_n, [1]_n, \ldots, [n-1]_n\}.$$

Operations are defined modulo n:

$$[a]_n + [b]_n = [a+b]_n, \qquad [a]_n[b]_n = [ab]_n.$$

A useful ring consequence

Theorem. Let R be any ring and let $x \in R$. Then

$$x0 = 0 = 0x.$$

Proof idea. Let $y = x0$. Then

$$y + y = x0 + x0 = x(0 + 0) = x0 = y.$$

Add $-y$ to both sides:

$$y = 0.$$

Thus $x0 = 0$. The proof of $0x = 0$ is similar.

Abstract algebra often works like this: axioms force useful consequences.

Definition of a field

A **field** is a commutative ring F with identity $1 \neq 0$ such that every non-zero element has a multiplicative inverse.

Equivalently, in a field we can add, subtract, multiply, and divide by non-zero elements.

Examples.

$$\mathbb{Q}, \qquad \mathbb{R}, \qquad \mathbb{C}, \qquad \mathbb{Z}_p \text{ when } p \text{ is prime.}$$

Non-examples.

$$\mathbb{Z}, \qquad \mathbb{Z}_6, \qquad \mathrm{M}_2(\mathbb{R}), \qquad \mathbb{Z}_n \quad \text{when } n \text{ is composite.}$$

Example: inverses in $\mathbb{Z}_5$

In $\mathbb{Z}_5$, every non-zero element has an inverse:

a	1	2	3	4
a^{-1}	1	3	2	4

Check:

$$2 \cdot 3 = 6 \equiv 1 \pmod 5, \qquad 4 \cdot 4 = 16 \equiv 1 \pmod 5.$$

In contrast, $\mathbb{Z}_6$ is not a field because $[2]_6$ has no multiplicative inverse.

A slightly larger field: $\mathbb{Q}[\sqrt{2}]$

Define

$$\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}.$$

This is a field. The main point is inverses. If $a + b\sqrt{2} \neq 0$, then

$$(a + b\sqrt{2})^{-1} = \frac{a - b\sqrt{2}}{a^2 - 2b^2}.$$

Indeed,

$$(a + b\sqrt{2})(a - b\sqrt{2}) = a^2 - 2b^2.$$

This illustrates enlarging a field by adjoining a new number.

Quick check: rings and fields

Decide whether each object is a ring, a field, both, or neither.

1. $\mathbb{Z}$ under usual $+$ and $\cdot$.
2. $2\mathbb{Z}$ under usual $+$ and $\cdot$.
3. $\mathbb{Z}_5$ under modular $+$ and $\cdot$.
4. $\mathbb{Z}_6$ under modular $+$ and $\cdot$.
5. $\mathrm{M}_2(\mathbb{R})$ under matrix $+$ and $\cdot$.
6. $\mathbb{Q}[\sqrt{2}]$ under usual $+$ and $\cdot$.

For each non-field, identify what fails.

Vector spaces

A first example: $\mathbb{R}^2$

Let

$$\mathbb{R}^2 = \{(x, y) : x, y \in \mathbb{R}\}.$$

We can add elements coordinatewise:

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2).$$

We can also multiply an element by a real number:

$$\lambda(x, y) = (\lambda x, \lambda y).$$

For example,

$$2(3, -1) = (6, -2).$$

Elements of $\mathbb{R}^2$ are called **vectors**; elements of $\mathbb{R}$ are called **scalars**.

What structure does $\mathbb{R}^2$ have?

The set $\mathbb{R}^2$ has two kinds of operations.

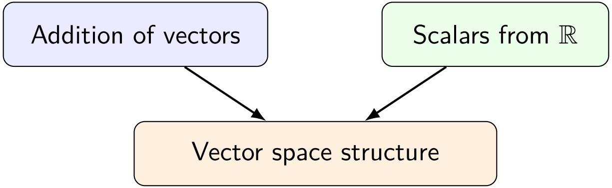
- We can **add vectors**:

$$u + v \in \mathbb{R}^2.$$

- We can **scale vectors** by real numbers:

$$\lambda v \in \mathbb{R}^2.$$

So $\mathbb{R}^2$ combines:



Definition of a vector space

Let F be a field.

A **vector space** over F is an additive Abelian group $(V, +)$ together with scalar multiplication

$$F \times V \to V, \qquad (\lambda, v) \mapsto \lambda v,$$

satisfying:

$$(\lambda\mu)v = \lambda(\mu v), \qquad 1v = v,$$
$$(\lambda + \mu)v = \lambda v + \mu v, \qquad \lambda(u + v) = \lambda u + \lambda v.$$

Elements of V are **vectors**; elements of F are **scalars**.

Examples of vector spaces

The example $\mathbb{R}^2$ is a vector space over $\mathbb{R}$.

More generally,

$$\mathbb{R}^n = \{(x_1, \ldots, x_n) : x_i \in \mathbb{R}\}$$

is a vector space over $\mathbb{R}$, with coordinatewise addition:

$$(x_1, \ldots, x_n) + (y_1, \ldots, y_n) = (x_1 + y_1, \ldots, x_n + y_n),$$

and coordinatewise scalar multiplication:

$$\lambda(x_1, \ldots, x_n) = (\lambda x_1, \ldots, \lambda x_n).$$

More examples

- $\mathbb{C}^n$ over $\mathbb{C}$.
- $\mathbb{C}$ over $\mathbb{R}$.
- $\mathbb{R}[x]$, the set of real polynomials, over $\mathbb{R}$.
- $\mathrm{M}_2(\mathbb{Z}_p)$ over $\mathbb{Z}_p$, where p is prime.

For matrices over $\mathbb{Z}_p$,

$$\lambda \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} \lambda a & \lambda b \\ \lambda c & \lambda d \end{pmatrix},$$

with entries computed modulo p.

Vector spaces over finite fields

Because $\mathbb{Z}_p$ is a field when p is prime, we can form vector spaces over $\mathbb{Z}_p$.

For example,

$$\mathbb{Z}_2^n = \{0,1\}^n$$

is a vector space over $\mathbb{Z}_2$.

Addition is coordinatewise modulo 2:

$$(1,0,1) + (1,1,0) = (0,1,1).$$

This viewpoint will be important later for linear error-correcting codes.

Subspaces

Let V be a vector space over a field F, and let W be a nonempty subset of V:

$$W \subseteq V.$$

We say that W is a **subspace** of V if W is a vector space using the same addition and scalar multiplication as V.

Equivalently, $W \neq \emptyset$ is a subspace if it satisfies the following two closure properties:

$$u, v \in W \Rightarrow u + v \in W, \qquad \lambda \in F,\ v \in W \Rightarrow \lambda v \in W.$$

Example in $\mathbb{R}^2$:

$$W = \{(t, 2t) : t \in \mathbb{R}\}$$

is a subspace, but

$$W' = \{(t, 2t+1) : t \in \mathbb{R}\}$$

is not, since it does not contain $(0, 0)$.